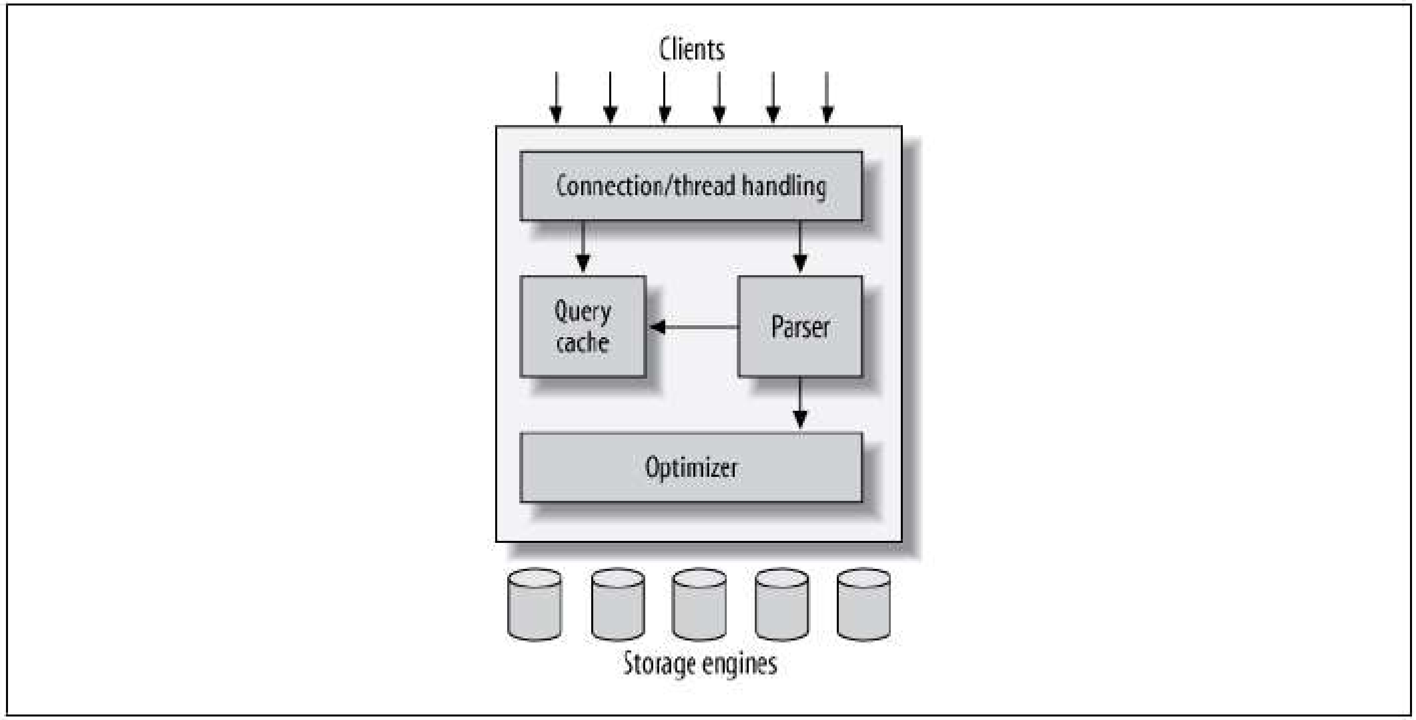
MySQL Architecture

Logical architecture



MySQL components and functions

- Connection management and security
- Optimization and execution
- Concurrency control
- Read/write locks - Table locks, Row locks
- Transactions - ACID test
- Deadlock
- Commit

Transactions

- Transferring money from Current account to Savings account
- START TRANSACTION;
- Update current_account set balance=balance – 200 where cust_id=124578;
- Update saving_account set balance = balance + 200 where cust_id=124578;
- COMMIT;

ACID

- Atomicity – All or nothing
- Consistency – Same data
- Isolation – Incomplete trx results not visible to others
- Durability – Committed data is permanent

Deadlocks

- Two transactions requesting for lock on same resource

Transaction #1

```
START TRANSACTION;
UPDATE StockPrice SET close = 45.50 WHERE stock_id = 4 and date = '2002-05-01';
UPDATE StockPrice SET close = 19.80 WHERE stock_id = 3 and date = '2002-05-02';
COMMIT;
```

Transaction #2

```
START TRANSACTION;
UPDATE StockPrice SET high  = 20.12 WHERE stock_id = 3 and date = '2002-05-02';
UPDATE StockPrice SET high  = 47.20 WHERE stock_id = 4 and date = '2002-05-01';
COMMIT;
```

Commit

- AUTOCOMMIT enabled by default in MySQL

```
mysql> SHOW VARIABLES LIKE 'AUTOCOMMIT';
+---------------+-------+
| Variable_name | Value |
+---------------+-------+
| autocommit    | ON    |
+---------------+-------+
1 row in set (0.00 sec)
mysql> SET AUTOCOMMIT = 1;
```